

DR. RAJIB BARAN ROY, PhD

Tier-1 International Postdoctoral Research Portfolio

Professional Identity

ORCID: 0000-0002-1177-0778

Google Scholar: <https://scholar.google.com/citations?hl=en&user=oowfLzIAAAAJ>

Research Areas: Electrical Power Systems, Smart Grids, Renewable Energy Integration, Electric Transportation, Battery Energy Storage Systems, AI for Energy Systems

Executive Research Profile

Electrical Power Systems researcher with nearly two decades of academic experience and an established international publication record. Research expertise spans smart grids, electric ferry charging infrastructure, distribution network analysis, battery energy storage systems, renewable energy integration, microgrids, and AI-driven optimization. Extensive experience with OpenDSS, DIgSILENT PowerFactory, PSCAD, MATLAB/Simulink, and Python-based power system modelling.

Research Impact

Google Scholar Citations: 492

Google Scholar h-index: 13

Google Scholar i10-index: 16

Scopus Citations: 356

Scopus h-index: 7

Academic Book: 1

Journal Publications: 23

Conference Publications: 19+

International Research Grants: 3

ATS-Optimized Research Keywords

Power Systems, Smart Grids, Distribution Networks, Renewable Energy Integration, Electric Vehicles,

Electric Ferry Charging, Battery Energy Storage Systems, Energy Management Systems, Microgrids, Artificial Intelligence, Machine Learning, Metaheuristic Optimization, Power Quality, Grid Stability, Digital Twin, Vehicle-to-Grid, Renewable Hydrogen, Power System Resilience, Decarbonization, Net-Zero Energy Systems

Education

PhD in Electrical Engineering, Central Queensland University, Australia (2022–2026)
MSc in Sustainable Energy Systems and Management, University of Flensburg, Germany (2006–2008)

Academic Appointments

Assistant Professor, Primeasia University (2016–Present)
Senior Lecturer, University of Information Technology and Sciences (2009–2016)
Lecturer, University of Information Technology and Sciences (2004–2006)

Top 5 Most Influential Publications

1. Impact of Coordinated Electric Ferry Charging on Distribution Network Using Metaheuristic Optimization, *Energies* (2025)
2. Impact Analysis on Distribution Network due to Coordinated Electric Ferry Charging, *IET Energy Systems Integration* (2024)
3. Optimizing Dynamic Ferry Loads with Intelligent Power Management, *Energy Reports* (2023)
4. Comparative Performance Analysis of ANN Algorithms for MPPT Energy Harvesting in Solar PV Systems, *IEEE Access* (2021)
5. Grid Impacts of Uncoordinated Fast Charging of Electric Ferries, Batteries (2021)

Research Funding and Industry Collaboration

Green Shore Power Project – Gladstone Ports Corporation, Australia
Queensland Reconstruction Authority – Resilience Energy Hub Project
Ministry of Higher Education Malaysia LRGS Project

Collaborative research involving academia, government agencies, and industry stakeholders in energy transition and infrastructure resilience.

Future Research Agenda

Research Theme 1: AI-Driven Smart Distribution Networks
Research Theme 2: Electric Transportation Infrastructure Optimization
Research Theme 3: Battery Energy Storage and Grid Flexibility
Research Theme 4: Digital Twins for Power Systems
Research Theme 5: Coastal and Remote Community Energy Systems
Research Theme 6: Renewable Hydrogen and Sector Coupling
Research Theme 7: Net-Zero and Climate-Resilient Energy Infrastructure

Long-Term Vision:

Develop intelligent, resilient, and sustainable energy systems capable of supporting large-scale renewable integration and transportation electrification.

Two-Page Research Statement

My research focuses on the intersection of power systems, renewable energy integration, electric transportation, and artificial intelligence. The primary objective is to develop intelligent and resilient electricity networks capable of supporting future decarbonized energy systems.

My doctoral research investigates the impacts of fast-charging electric ferries on distribution networks and develops optimization strategies for charging coordination, energy storage integration, and network stability improvement. This work contributes to the emerging field of maritime electrification and sustainable transportation.

Future research will extend these contributions through AI-enabled decision support systems, digital twins, battery energy storage optimization, renewable hydrogen integration, and resilient coastal energy systems. The overarching goal is to enable reliable, economically viable, and environmentally sustainable electricity infrastructure for future communities.

The research methodology combines advanced power system modelling, optimization algorithms, machine learning, data analytics, and industry collaboration. Key software platforms include OpenDSS, DIgSILENT PowerFactory, PSCAD, MATLAB/Simulink, and Python.

Through international collaboration and interdisciplinary research, I aim to contribute to global energy transition initiatives and support the development of next-generation smart energy systems.

Master Postdoctoral Cover Letter

Dear Selection Committee,

I am applying for the Postdoctoral Research Fellow position in Electrical Engineering, Energy Systems, Smart Grids, or Renewable Energy. I am completing my PhD in Electrical Engineering at Central Queensland University, Australia, with research focused on electric ferry charging, distribution network impacts, battery energy storage systems, and optimization techniques.

My publication record includes 23 journal papers, 19+ conference papers, one academic book, and more than 492 Google Scholar citations. My research has appeared in leading journals including Energies, IEEE Access, Energy Reports, Batteries, Micromachines, and IET

Energy Systems Integration.

I have extensive experience using OpenDSS, DigSILENT PowerFactory, MATLAB/Simulink, PSCAD, and Python for power system analysis and optimization. In addition, I have participated in industry-funded and government-funded projects involving energy resilience, smart grids, and sustainable transportation.

I would welcome the opportunity to contribute to internationally recognized research programs addressing the challenges of energy transition, grid modernization, and transportation electrification.

Sincerely,

Dr. Rajib Baran Roy